



National University

Of Computer & Emerging Sciences Faisalabad-Chiniot Campus

AI 2002 Artificial Intelligence

Course Instructor

Ms. Mahzaib Younas

Time allowed = 40 min

Quiz 2

Total Marks = 35

BCS Section E

Roll No _____

Name _____

Signature _____

Question No 01: Choose the correct one.

[6]

1. Which of the following statements about heuristic functions is FALSE? a) An admissible heuristic never overestimates the true cost to the goal. b) The Manhattan distance heuristic can be used for the 8-puzzle problem. c) Consistent heuristic ensures that estimated cost of a path does not decrease. d) Heuristic that is admissible but not consistent can guarantee optimality in A*.	2. In which scenario would Uniform-Cost Search perform worse than Breadth-First Search? a) When path costs are significantly varied b) When the search tree has cycles c) When the step costs are uniform d) When the heuristic information is available
3. Which of the following search algorithms is guaranteed to find an optimal solution, provided that all step costs are positive and the heuristic is admissible? a) Depth-First Search b) Greedy Best-First Search c) Iterative Deepening Depth-First Search d) A* Search	4. Which search algorithm is best suited for solving problems with infinite state spaces and avoids redundant state exploration? a) Depth-First Search b) Breadth-First Search c) Uniform-Cost Search d) Iterative Deepening Search
5. Which of the following search strategies expands the node with the lowest path cost first? a) A* Search b) Best-First Search c) Greedy Best-First Search d) Uniform-Cost Search	6. At each iteration of its main loop, A* needs to determine which part of it need to extend. a) Paths b) Node c) Route d) Heuristic function

Question no 02: Calculate the heuristic of the given problem.

[3 Marks]

4	8	6
5		2
7	1	3

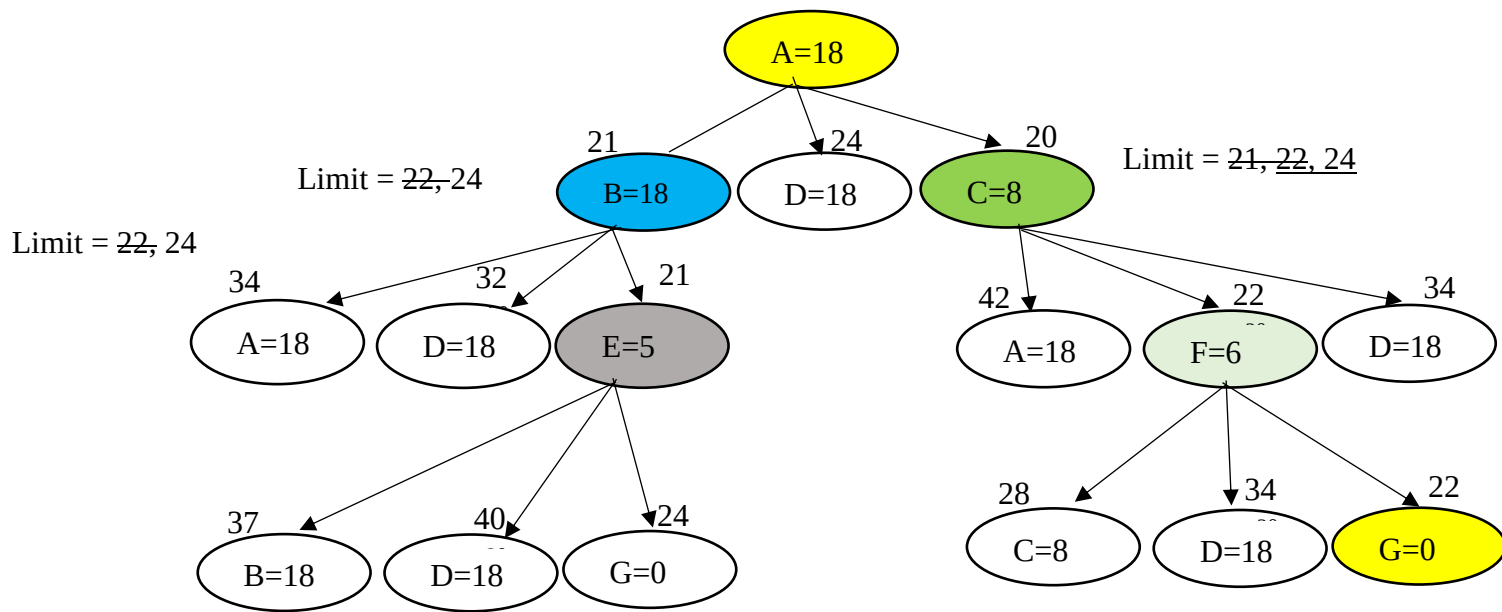
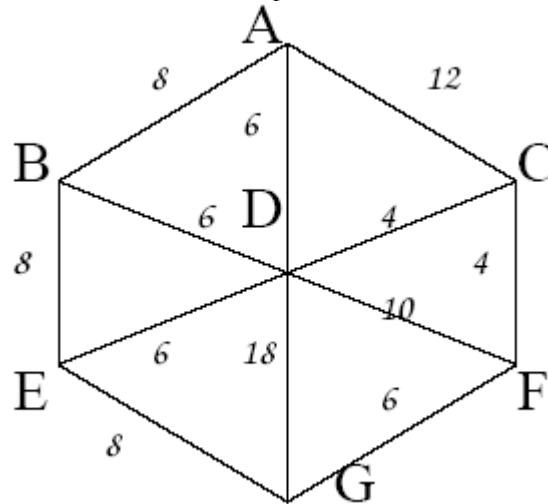
1	2	3
8		4
7	6	5

1	3
2	2
3	2
4	3
5	3
6	3
7	0

| Initial State | Goal State | 8 | 2 |

Question No 03: The numbers in the graph below are the real distances between the nodes. The estimated distances to the goal node, G, are the following: **A: 18, B: 13, C: 8, D: 18, E: 5, F: 6, G:0**

A is the start node. Implement a search with algorithm recursive best first search using graph search. Show how this algorithm works. Mark also in the tree the order in which the nodes are expanded. **[8 Marks]**



Final Path = ACFG
Total Cost = 22

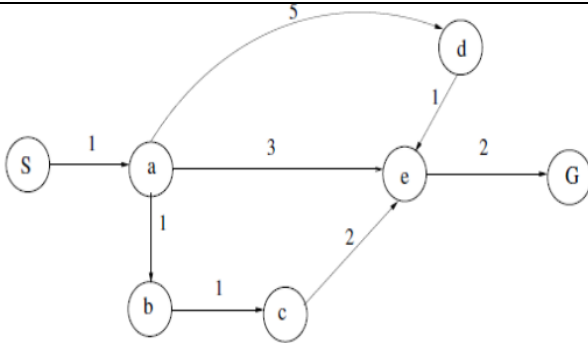
Question No 03: Consider the following map, start is S and goal is G.

[10 Marks]



National University

Of Computer & Emerging Sciences Faisalabad-Chiniot Campus



Node	Value	Updated heuristics
S	6	6
a	6	5
b	5	5
c	3	4
d	2	2
e	1	2
G	0	0

Check whether the given heuristic is consistent or not. If not consistent make it consistent.

Node	Consistency			
S	$S \rightarrow a, 6 - 6 \leq 1, 0 \leq 1$		$S \rightarrow a, 6 - 6 \leq 1, 0 \leq 1$	$S \rightarrow a, 6 - 6 \leq 1, 0 \leq 1$
a	$a \rightarrow d, 6 - 2 \leq 5, 4 \leq 5$ $a \rightarrow e, 6 - 1 \leq 3, 5 \leq 3$ $a \rightarrow b, 6 - 5 \leq 1, 1 \leq 1$	H(e) = 3	$a \rightarrow d, 6 - 2 \leq 5, 4 \leq 5$ $a \rightarrow e, 6 - 3 \leq 3, 3 \leq 3$ $a \rightarrow b, 6 - 5 \leq 1, 1 \leq 1$	$H(a) = 5$ $a \rightarrow d, 5 - 2 \leq 5, 3 \leq 5$ $a \rightarrow e, 5 - 2 \leq 3, 3 \leq 3$ $a \rightarrow b, 5 - 5 \leq 1, 0 \leq 1$
B	$b \rightarrow c, 5 - 3 \leq 1, 2 \leq 1$		$b \rightarrow c, 5 - 4 \leq 1, 1 \leq 1$	$b \rightarrow c, 5 - 3 \leq 1, 2 \leq 1$
C	$c \rightarrow e, 3 - 1 \leq 2, 2 \leq 2$	H(c) = 4	$c \rightarrow e, 4 - 3 \leq 2, 1 \leq 2$	$c \rightarrow e, 3 - 2 \leq 2, 1 \leq 2$
d	$d \rightarrow e, 2 - 1 \leq 1, 1 \leq 1$		$d \rightarrow e, 2 - 3 \leq -1, -1 \leq 1$	$d \rightarrow e, 2 - 2 \leq 0, 0 \leq 1$
e	$e \rightarrow G, 1 - 0 \leq 2, 1 \leq 2$		$e \rightarrow G, 3 - 0 \leq 2, 1 \leq 2$	$H(e) = 2$ $e \rightarrow G, 2 - 0 \leq 2, 2 \leq 2$
G				

Apply the A* after updated heuristics with expanded list.

Node	Expanded list
(0 S)	<u>S</u>
(6 A S)	<u>AS</u>
(6 EAS) (8 DAS) (7BAS)	<u>EAS</u>
(6GEAS) (8DAS) (7BAS)	<u>GEAS</u>

Part c: Consider three heuristics h1, h2, h3. The table below indicates the estimated cost to goal (h value) for each of the heuristics for each node in the search graph [4 Marks]

node	h_1	h_2
S	6	6
a	5	5
b	5	4
c	4	2
d	2	1
e	2	1
G	0	0

```
graph LR; S((S)) -- 1 --> a((a)); a -- 1 --> b((b)); b -- 1 --> c((c)); c -- 2 --> e((e)); a -- 3 --> e; e -- 1 --> d((d)); d -- 5 --> a; e -- 2 --> G((G))
```

For each heuristic function, circle whether it is admissible and whether it is consistent with respect to the search problem given above.

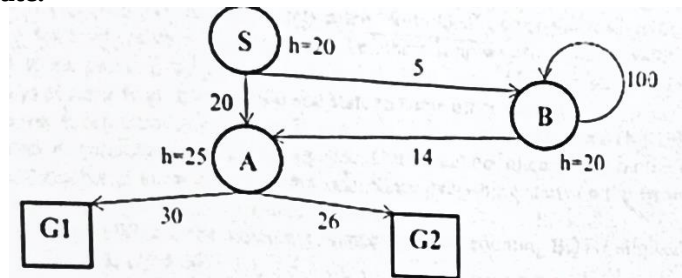


National University

Of Computer & Emerging Sciences Faisalabad-Chiniot Campus

heuristics	admissible		Consistence	
H1	Yes	No	Yes	No
H2	Yes	No	Yes	No

Question No 04: consider the following graph and expanded the it in left to right order where S is the start state and G1 and G2 is the goal state. [4 Marks]



Uniform Cost Search:

Order of expansion: SBAG₂
Path: SBAG₂
Total Cost: 45

Best first Search:

Order of expansion: S,B,B,B,.....
Path: No Path
Total Cost: Infinite loop